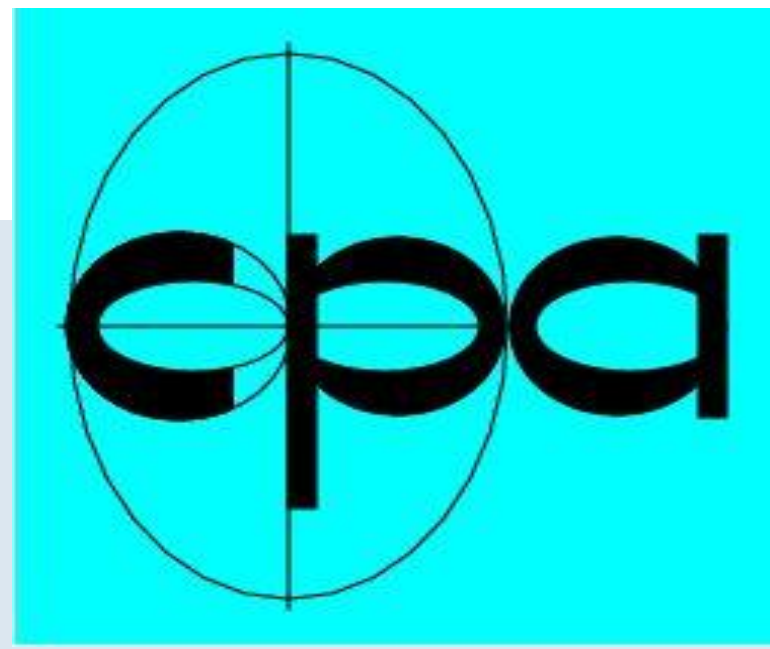




Probabilistic Solar Flare Forecasting with Weakly Supervised Contrastive Learning

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About me:



Predicting Solar Eruptions

A prove that we understand the underlying processes and can identify precursors

Mitigation of the adverse space weather effects

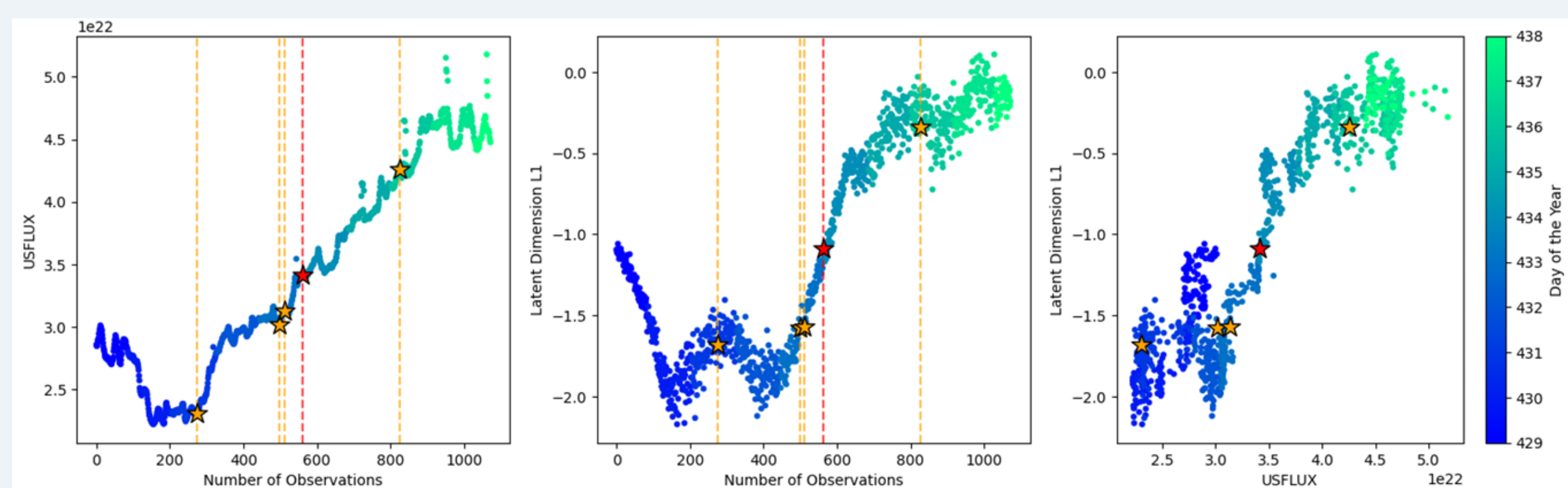
Experienced **observers** can often assess the flare potential of an active region by following its evolution over time → combining visual cues with domain knowledge.

ML-supported forecasting can mirror this approach → data-driven features + physics-based constraints

- **Representation learning**: compress high-dimensional magnetic field observation into a low-dimensional, **informative latent description**
- **Disentangled factors**: latent variables to **capture distinct aspects of magnetic morphology**, supporting interpretability
- **Contrastive learning**: use domain-informed similarity rules to **organize the latent space**

β-VAE: Latent Space and Active Region Parametrization

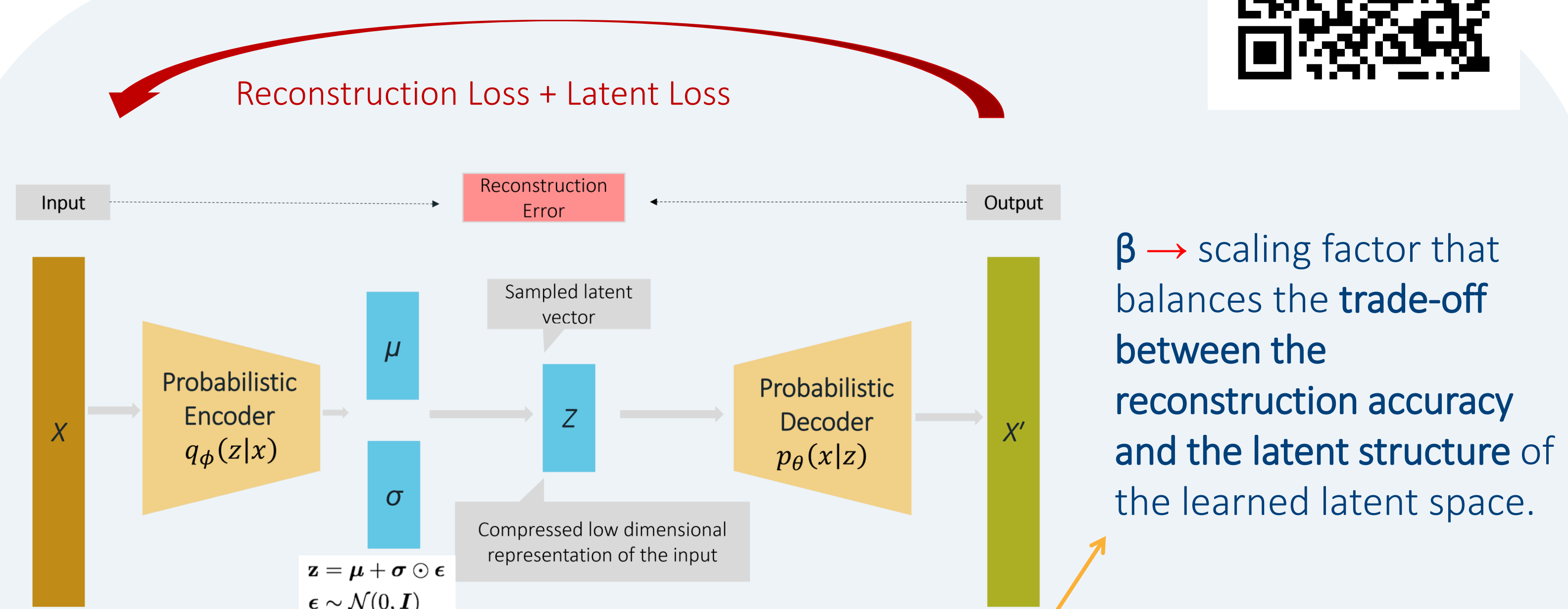
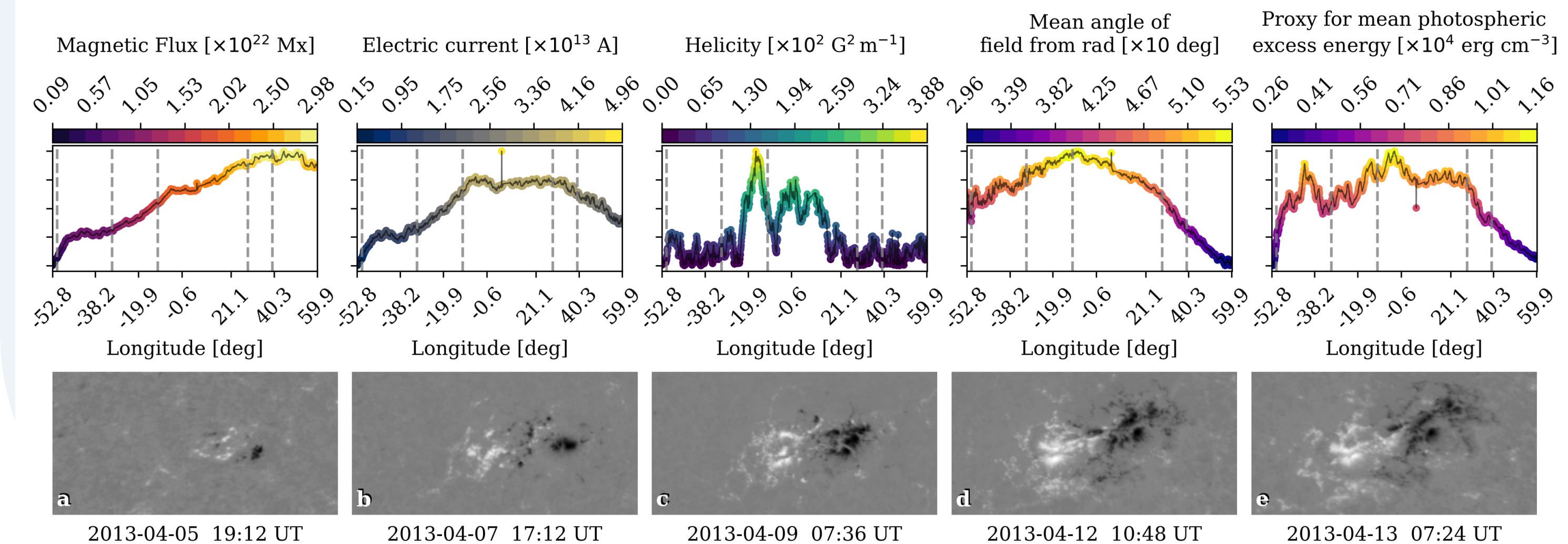
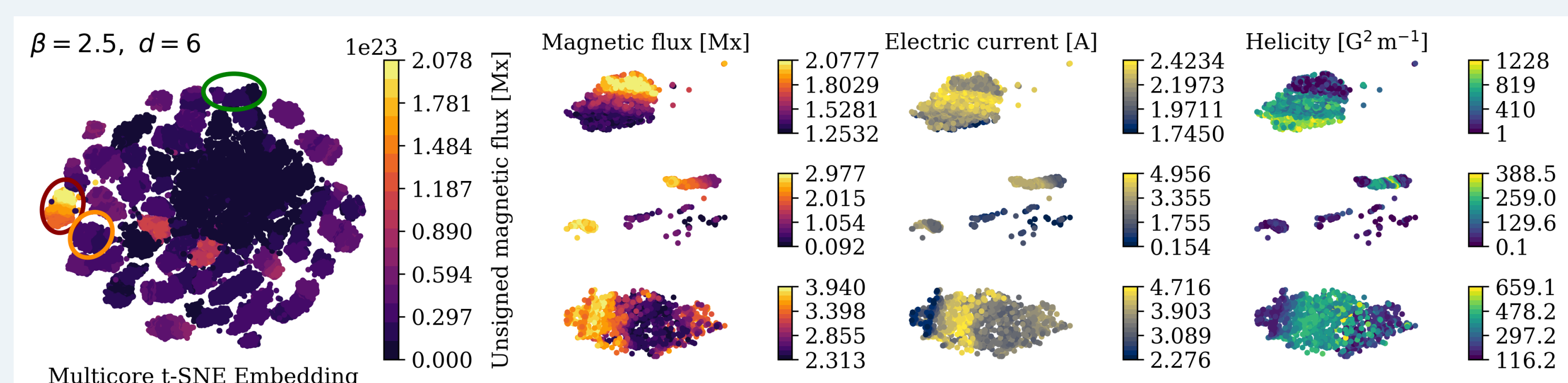
- Linear relation between empirical and latent features helps us “explain” the latter
- Features are mostly related to change in **size, dynamics, and magnetic morphology**
- **location and tilt** → cycle properties related to dynamo action



Large structure → a slightly off-centered core where **low-flux** active regions are concentrated and **enhanced-flux** clusters halo.

Individual clusters:

- observations with similar magnetic properties tend to lie close together, even though the embedding is based on only 6 latent features.
- middle active region times series (22 flares) but lower overall flux distribution → closer to the core
- Higher flux, large size → more compact clusters; less fractioning



$$\mathcal{L}(\theta, \phi; \mathbf{x}, \mathbf{z}, \beta) = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p_{\theta}(\mathbf{x}|\mathbf{z})] - \beta D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$$

□ **Triplet-trained, Siamese-like**: three inputs - an anchor, positive, and a negative frame are processed by identical, weight-sharing β-VAE encoders.

□ **Goal**: Learn a physically meaningful latent space in which ***B_r*** **maps** with similar **activity-related properties** cluster together and dissimilar ones remain separated.

□ **How**: A triplet loss is added to the β-VAE reconstruction and KL-divergence losses, pulling the anchor toward the positive and pushing it away from the negative

Step 1: Chronological split including active region number safeguard

Step 2: Within each training subset, assign **SHARP-threshold** labels: high-activity, low-activity, or ambiguous.

Step 3: Keep only the clear **high- and low-activity** samples for training (for now)

Step 4: Build triplets separately in each subset: anchor, same-class positive, opposite-class negative.

Step 5: Train the Siamese β-VAE on the training triplets and monitor performance on the validation triplets

Step 6: Use the held-out test subset only for final **evaluation and preliminary forecast/latent space assessment**.

